$plot(\sin(\pi * x) - x^3 \cdot (-16 \cdot \sqrt{2} + 4 \cdot \pi \cdot \sqrt{2}))$

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Интерполяционный полином Эрмита

$f(x) = \sin(\pi x)$; $x \in [-\frac{1}{4}; \frac{1}{4}]$; $n = 1$

$\frac{df}{dx} = \pi\cos(\pi x)$

№	x	$f(x)$	$\frac{df}{dx} = m$
0	$-\frac{1}{4}$	$-\frac{\sqrt{2}}{2}$	$\frac{\pi\sqrt{2}}{2}$
1	$\frac{1}{4}$	$\frac{\sqrt{2}}{2}$	$\frac{\pi\sqrt{2}}{2}$

$h_1 = x_1 - x_0 = \frac{1}{4} + \frac{1}{4} = \frac{1}{2}$

$t = \frac{x - x_0}{h_1} = \frac{x + \frac{1}{4}}{\frac{1}{2}} = \frac{4x+1}{2}$

$H_1(t) = f_0(1 - 3t^2 + 2t^3) + f_1(3t^2 - 2t^3) + m_0 h_1(t - 2t^2 + t^3) + m_1 h_1(-t^2 + t^3)$

$H_1(t) = -\frac{\sqrt{2}}{2}(1 - 3t^2 + 2t^3) + \frac{\sqrt{2}}{2}(3t^2 - 2t^3) + \frac{\pi\sqrt{2}}{2}\frac{1}{2}(t - 2t^2 + t^3) + \frac{\pi\sqrt{2}}{2}\frac{1}{2}(-t^2 + t^3) =$

$= \frac{\sqrt{2}(6t^2 - 4t^3 - 1)}{2} + \frac{\pi\sqrt{2}(2t^3 - 3t^2 + t)}{4} = \frac{6t^2 - 4t^3 - 1}{\sqrt{2}} + \frac{\pi(2t^3 - 3t^2 + t)}{2\sqrt{2}}$

$H_1(x) = \frac{6(2x+\frac{1}{2})^2 - 4(2x+\frac{1}{2})^3 - 1}{\sqrt{2}} + \frac{\pi(2(2x+\frac{1}{2})^3 - 3(2x+\frac{1}{2})^2 + 2x + \frac{1}{2})}{2\sqrt{2}} = \sqrt{2}(3x - 16x^3) +$

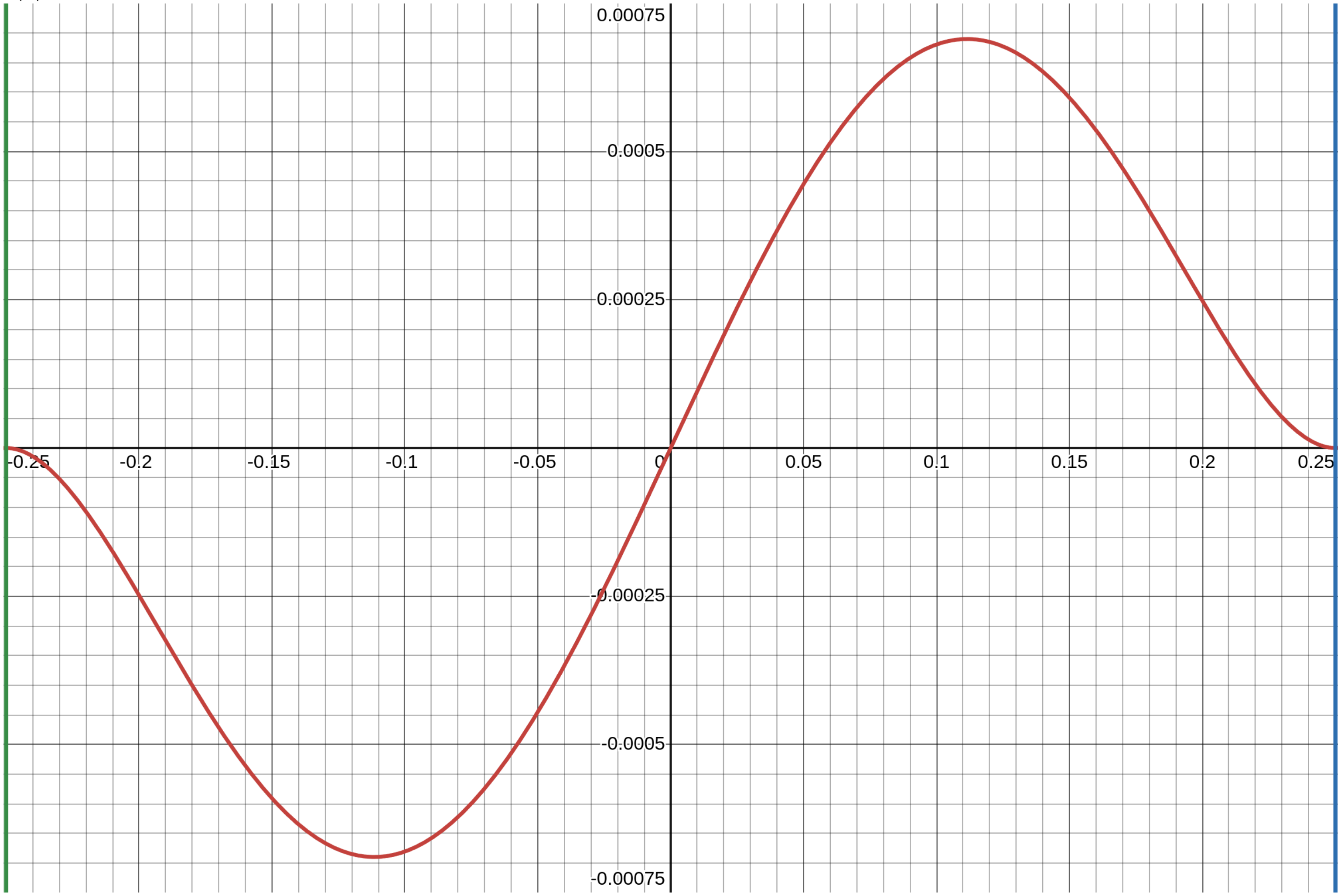
$+ \frac{\pi(16x^3 - x)}{2\sqrt{2}} = x^3(-16\sqrt{2} + 4\pi\sqrt{2}) + x(3\sqrt{2} - \frac{\pi\sqrt{2}x}{4})$

График погрешности: $H_1(x) = T_1(x)$

$r(x) = f(x) - T_1(x) = \sin(\pi x) - x^3(-16\sqrt{2} + 4\pi\sqrt{2}) - x(3\sqrt{2} - \frac{\pi\sqrt{2}}{4})$

$\frac{dr(x)}{dx} = \pi\cos(\pi x) - 3x^2(-16\sqrt{2} + 4\pi\sqrt{2}) - (3\sqrt{2} - \frac{\pi\sqrt{2}}{4})$

$r(x)$: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0.0005
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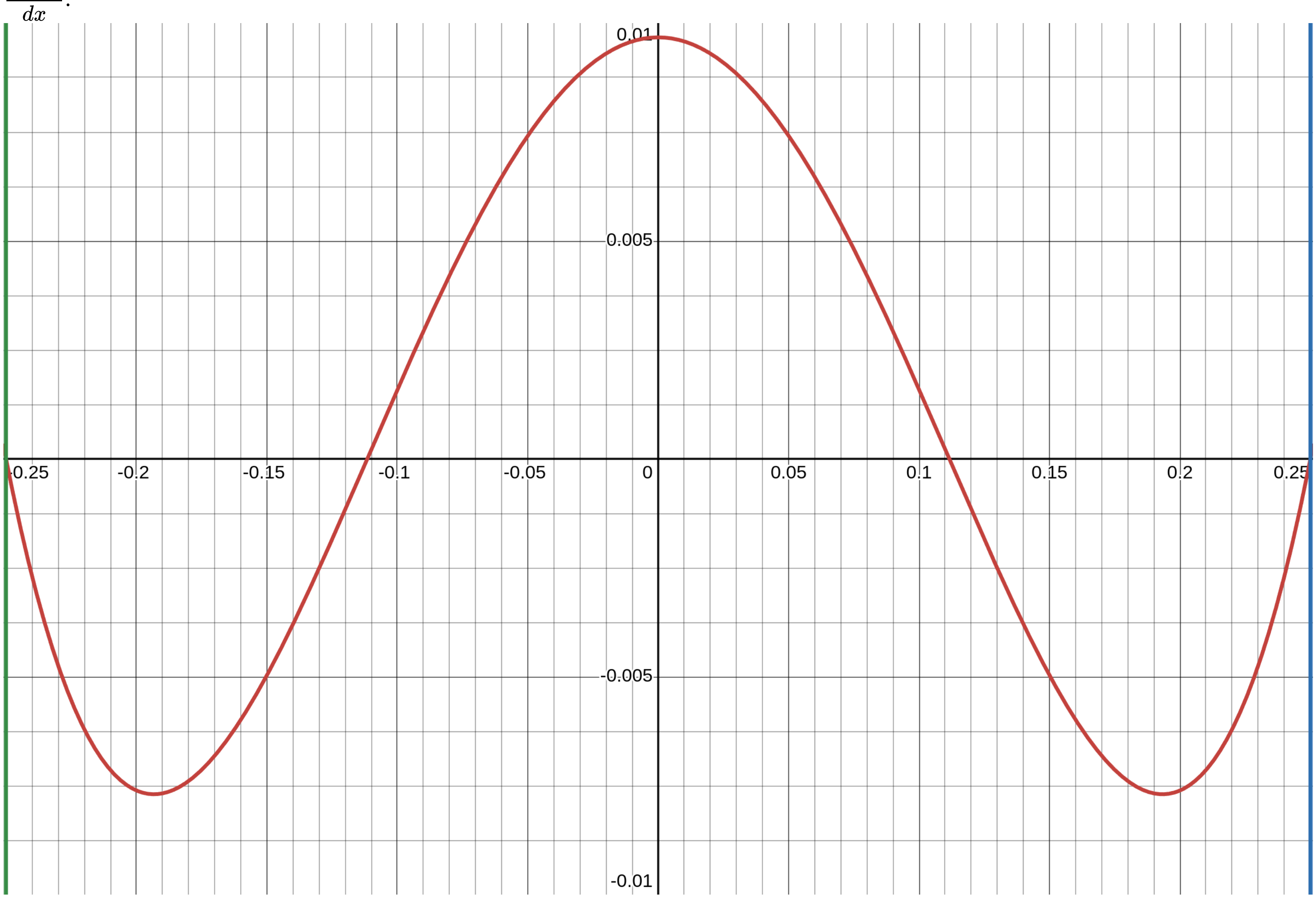
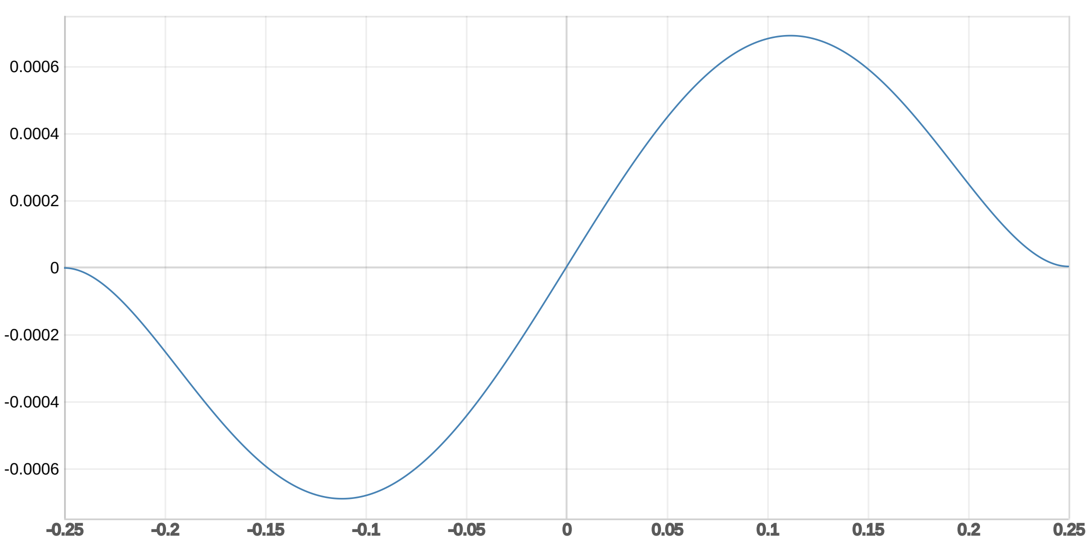
-0.0005
-0.00075
-0.25
-0.2
-0.15
-0.1
-0.05
0
0.05
0.1
0.15
0.2
0.25

$\frac{dr(x)}{dx}$:



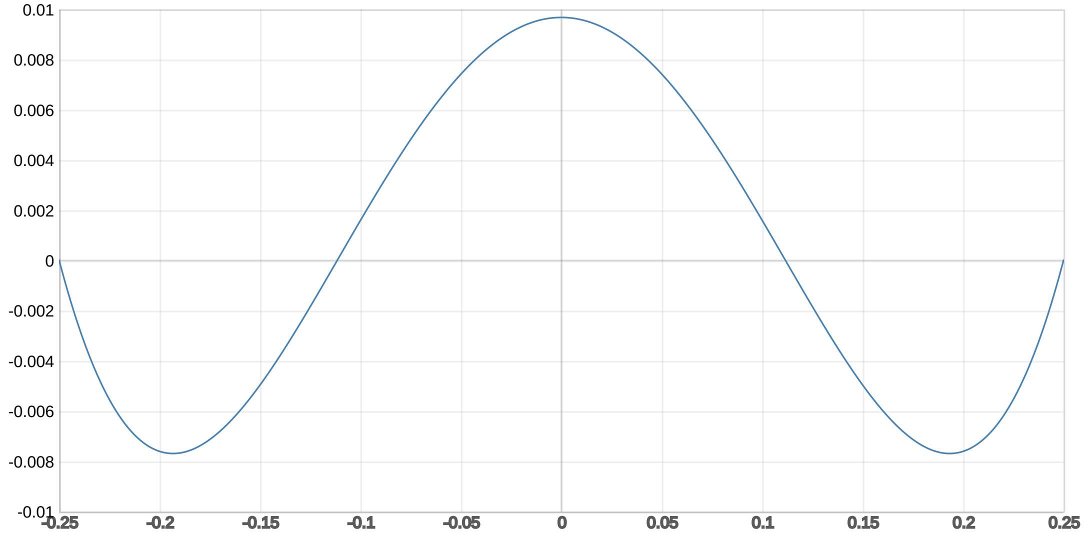
$r(x)$:

$$[4.8559 \cdot x^3 - 3.13191 \cdot x + \sin(3.141593 \cdot x)]$$



$\frac{dr(x)}{dx}$:

$$[3.141593 \cdot \cos(3.141593 \cdot x) + 14.5677 \cdot x^2 - 3.13191]$$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0.006
0.004
0.002
0
-0.002
-0.004
-0.006
-0.008
-0.01
-0.25
-0.2
-0.15
-0.1
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0
0.05
0.1
0.15
0.2
0.25